LISINOPRIL

Once daily
Teratogenic

CHEMICAL NAME (2S)-1-[(2S)-6-amino-2-{[(1S)-1-carboxy-3-phenylpropyl]amino}hexanoyl]pyrrolidine-2-carboxylic acid (dihydrate)
CLASS ACE inhibitor (long-acting, non-prodrug)
TRADE NAMES Zestril · Prinivil · Lisoril · Zestoretic (with HCTZ)

THERAPEUTIC USES / INDICATIONS

- Hypertension.
- Heart failure.
- Acute myocardial infarction (within 24 h, haemodynamically stable).
- Diabetic nephropathy.

ADMINISTRATION

- **PO** once daily at the same time, with or without food.
- Not a prodrug — no hepatic activation needed.

DOSAGES

- **Hypertension:** start 10 mg daily (2.5–5 mg if on diuretic or renal impairment); usual 20–40 mg; max 80 mg.
- **Heart failure:** 2.5–5 mg daily, titrate to 20–35 mg.
- **Post-MI:** 5 mg within 24 h, 5 mg at 24 h, 10 mg at 48 h, then 10 mg daily.
- **Children ≥6 yr:** 0.07 mg/kg daily (max 5 mg start).

PHARMACOKINETICS

- **Onset:** 1 h. **Peak:** 6–8 h. **Duration:** 24 h.
- Not metabolised.
- Excreted unchanged in urine. **t½:** 12 h.

PHARMACODYNAMICS

- Inhibits **ACE** → ↓ angiotensin II → vasodilation, ↓ aldosterone.
- Renoprotective; reduces LV remodelling post-MI.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Cough, dizziness, headache.
- Fatigue, nausea.
- Hypotension.

ADVERSE EFFECTS (SERIOUS)

- Angioedema (higher risk in Black patients).
- Hyperkalaemia.
- Renal impairment.
- Pancreatitis, hepatitis (rare).

CONTRAINDICATIONS & CAUTIONS

- Pregnancy.
- Angioedema history.
- Bilateral renal artery stenosis; aortic stenosis (caution).
- Sacubitril/valsartan within 36 h.

KEY INTERACTIONS

- **K^+-sparing diuretics, supplements:** hyperkalaemia.
- **NSAIDs:** AKI.
- **Lithium:** toxicity.
- **Aliskiren:** avoid in diabetes.

NURSING MANAGEMENT

- Measure BP before dose; postural BP in elderly.
- Monitor U&E.
- Post-MI: monitor BP closely during first days.
- Report swelling of face/tongue.
- Daily weight in heart failure.

PATIENT EDUCATION

- Take once daily at the same time.
- Report swelling, fainting or persistent cough.
- Avoid potassium supplements unless prescribed.
- Use contraception; stop if pregnant and see doctor.

KEY POINTS

- Once-daily ACEi; not a prodrug.
- Cough, angioedema, hyperkalaemia.
- Renally excreted.
- Avoid in pregnancy.

SOURCES CHECKED [270] Lisinopril 1 mg/ml Oral Solution (UK SmPC) [271] lisinopril, oral (US label) [272] Lisinopril, oral (US label)
Full references p. 432–457